Build a red-line stopping station

Name: ____________________ Team: ____________________ Date: __________

A white approach lane with a broad red finish strip.

Gather these materials

[ ] Checked base with downward-facing color sensor

[ ] The tested white and red paper from lesson 4

[ ] Removable tape and ruler

[ ] Connected device with lesson 6 code

Build: steps 1-4

[ ] 1. Make a flat white lane at least twice as wide as the robot. Tape its edges down so the wheels cannot catch them. Keep tape away from the sensor’s reading path.

[ ] 2. Place a red strip across the full lane, starting about 10 cm deep in the direction of travel. This is a starting test size, not a guaranteed stopping distance.

[ ] 3. Leave a large clear area beyond the red strip. Mark a start close enough that a short drive can reach it. Ask the teacher to check the whole layout.

[ ] 4. With the wheels still, show the sensor white, red and another color. Fix wrong readings before driving. Keep the sensor gap from lesson 4.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[ ] 5. Build the program below. A variable is a value the program remembers. keepGoing remembers whether this test may continue.

[ ] 6. At low speed, try one approach to red. Keep someone ready at Stop. When the program ends, measure how far the sensor moved past the start of the red strip.

[ ] 7. Repeat three approaches. Separately test an unknown or other-color reading with wheels raised. After a red or unknown reading, show white again: the same test must stay stopped.

[ ] 8. If it misses red, stop. Try a wider strip or a lower speed, one change at a time. Do not remove the 20-check limit or the Stop steps.

Make the program

Start stopped. Set keepGoing to Yes.

Repeat at most 20 times. Only if keepGoing is Yes: read the color.

If white: move slowly for 0.1 seconds, then stop. Otherwise: stop and set keepGoing to No.

After the repeats, stop and end. A person must start a new test.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Saw red?	Distance past line (cm)	Stayed stopped?
1			
2			
3			

A successful build

Three approach runs stop when red is read.

Red and unknown both end the test; white cannot restart it.

The program ends after no more than 20 checks.

What could make the robot miss a narrow red strip?